



AR, VR AND THE METAVERSE (EEEM067) – SEM1 2025/6

Coursework Assessment:

“Beer Pong VR”

Instructors:

Dr. Marco Volino & Dr. Jean-Yves Guillemaut

1. Introduction and Concept

This project presents **Beer Pong VR**, an interactive virtual reality experience designed and implemented using **Unity** and the **Meta Quest 2** headset. The goal of the application is to recreate the mechanics of a real-world beer pong challenge within a structured, game-like environment. The player can move around a virtual room, pick up a ping pong ball, aim using a visual trajectory guide, and attempt to score by throwing the ball into a set of cups on a table. The experience includes a full gameplay loop consisting of scoring, timing, win/loss conditions, and the ability to restart the round.

The purpose of the application is to combine several core VR techniques from the laboratory sessions such as locomotion, interaction, UI integration, and physics simulation into a cohesive, polished, and engaging VR experience.

2. Key Features and Lab Topics Used

2.1 Interaction and Physics-Based Throwing (00:32–01:16)

The central mechanic of the game involves grabbing and throwing a ball. This feature integrates the **XR Interaction Toolkit**, **XR Grab Interactables**, and Unity **physics**. The player can pick up the ball, view a **trajectory prediction line**, and use the **A button** to launch it with a scripted velocity.

Lab Topics: Physical interaction, scripting, XR Grab Interactable, Rigidbody physics.

2.2 Teleportation Locomotion (01:17–00:30)

The player moves around the room using **teleportation**, implemented through Teleportation Areas and the XR Locomotion system. A custom mechanic also allows the player to return instantly to the starting position by pressing the **B button**.

Lab Topics: Locomotion systems, teleportation, controller input.

2.3 World-Space User Interface (00:29–00:34, 01:33–01:39)

The application includes a **floating world-space UI menu** with a Start button, Retry button, and dynamic header text ("Beer Pong", "You Win", "You Lose"). A HUD displays the **score** and **timer** during gameplay.

Lab Topics: VR UI design, world-space canvases, XR UI Input module.

2.4 Gameplay System: Score, Timer, Win/Lose Logic (00:32–01:17)

A custom **GameManager** controls the game flow. When the ball enters a cup, a trigger script registers the score, hides the cup, and resets the ball. A countdown timer creates pacing, and the game ends with either a **win condition** (all cups cleared) or **loss condition** (timer reaches zero).

Lab Topics: Scripting and game logic, event handling, triggers.

2.5 Rendering and Environment Design (01:17–01:31)

The room environment was enhanced with custom **materials** and **textures** on the floor, walls, and ping pong table. This improves realism and aesthetic quality while maintaining performance for standalone VR hardware.

Lab Topics: Materials, textures, scene organisation.

3. Timestamps Summary

Timestamp	Feature Demonstrated	Lab Topic
00:04–00:26	Main Menu (Beer Pong title, buttons)	VR UI
00:29–00:32	Start game	Scripting & UI
01:17–01:30	Teleportation movement	Locomotion
00:33–01:00	Grabbing ball + trajectory guide	Interaction & Physics
00:50–01:00	A-button throw	Controller input & physics
1:40–01:52	Cup scoring + ball respawn	Triggers & game logic
01:21–01:30	B-button teleport back	Locomotion & scripting
02:08–02:16	Clearing all cups → You Win	GameManager logic
01:36–01:41	Retry button resets the game	UI & logic
00:10–00:19	Environment / textures	Rendering

4. Technical Challenges and Solutions

4.1 Physics-Based Throwing Behaviour

One of the main challenges was ensuring that the ball launched forward accurately, following the projected trajectory. Unity's XR Interaction Toolkit applied velocity limits that caused the ball to stop mid-air. This was resolved by disabling velocity damping, controlling rigidbody physics fully in script, and applying a custom velocity vector.

4.2 Automatic Ball Respawn System

During development, the ball often fell off the table and rolled far away, making it difficult to retrieve. This was resolved by implementing a custom reset system that automatically returns the ball to its starting position whenever it touches the floor or goes out of bounds, ensuring smooth and consistent gameplay.

4.3 Teleportation and View Alignment

At times the XR Origin camera reset to the wrong position. To fix this, the [StartGame](#) method forcibly teleports the player to a [StartPosition](#) transform, ensuring consistent gameplay orientation. A [B-button](#) script allows recovery from accidental teleports.

4.4 User Interface Visibility and Event Handling

Ensuring that the UI appeared correctly in VR required configuration of the world-space canvas, scaling adjustments, and linking button events to the correct [GameManager](#) instance.

5. Conclusion

The Beer Pong VR application successfully integrates multiple VR development techniques taught in the module. It demonstrates physics-based interaction, teleportation locomotion, world-space UI, scripting, and environment design within a cohesive gameplay experience. The combination of interactive mechanics, clear UI, and polished environment results in an immersive VR mini-game that meets the coursework requirements and showcases both technical competence and creativity.